**4** This question is about an experiment to estimate the resolution of the human eye.

Two vertical parallel lines are drawn on a piece of card. The separation between the lines $y$ is 2.0 ± 0.5 mm. The card is fixed to a wall at head level.

A group of students look at the card whilst each covering one eye. They walk back from the card until they can no longer separate the two lines. The distance $L$ between the eye and the card is measured.

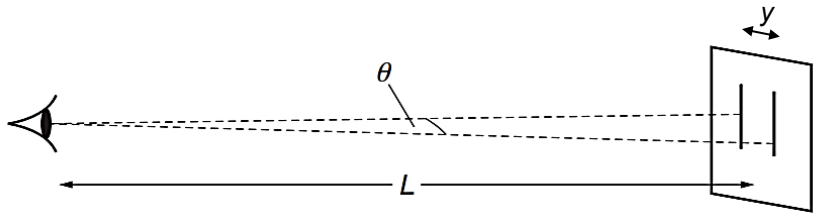


**Fig 4.1**

Here are the results from the five students:

| **student** | A | B | C | D | E |
|---|---|---|---|---|---|
| **maximum distance *L* / m** | 6.2 | 5.8 | 6.1 | 5.9 | 6.1 |

**(a)** **(i)** State the uncertainty of the distance $L$ based on students' results.

uncertainty = ..... m [1]

**(ii)** A student suggests that the uncertainty in the distance $L$ can be ignored when calculating the minimum angle $\theta$ that can be resolved because of the uncertainty in the separation of the lines $y$ on the card.

Comment on this suggestion, explaining whether or not you agree.

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..... [3]

**(iii)** Calculate a value for the minimum angle (with its corresponding uncertainty) that can be resolved.

minimum angle that can be resolved = …….…….. ± ……..……. rad [3]

**(b)** The diameter of the pupil of the eye is estimated to be 3 mm. The two lines are marked using red ink.

**(i)** Explain what is meant by the Rayleigh criterion for the resolution of two patterns.

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…… [2]

**(ii)** Estimate the minimum angle using the Rayleigh criterion.

minimum angle = …… rad [2]